

Ultrasonic Range

Time an Echo; publish a qualified reading

TP-E and three LEDs explain the circuit without Serial.

Lesson	019 — component	Time	120–180 minutes
Prerequisites	Lessons 001–018	Board	Arduino Mega 2560
Evidence	Echo/TP-E and D45–D47	Status	host verified; bench open
Safety	E1: USB-powered low-voltage circuit only		

1 Mission

Build an evidence chain:

Trigger → Echo width → `RangeState` → visible classification.

The ranger never turns “no pulse” into zero distance. A complete Echo outside the configured interval is `OutOfRange`; no accepted Echo is `Timeout`; only `Valid` carries a usable distance.

Boundary. This sensor is a learning instrument, not a collision, presence, occupancy, or personnel-safety device. Target shape, angle, material, air conditions, and module variation affect the result.

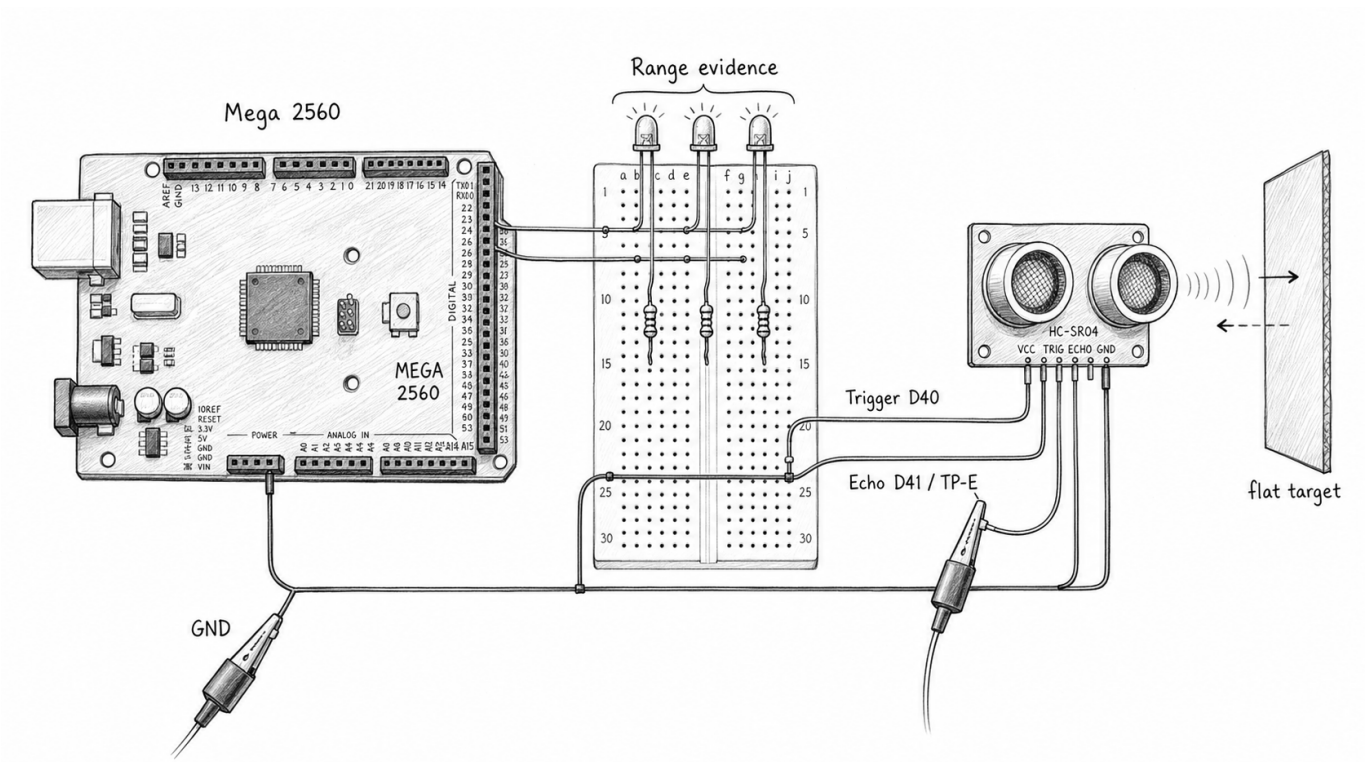
2 Parts, limits, and wiring

Part	Requirement
Mega 2560	USB powered, 5 V logic
Ultrasonic module	Exact identified Elegoo EL-SM-001, or bench work blocked
Evidence LEDs	Three LEDs, each with its own 220 Ω or larger resistor
Target	Flat cardboard, initially perpendicular to the sensor
Instrument	High-impedance logic probe or oscilloscope

Remove USB power before wiring or moving probe clips. Stop for a hot part, unstable USB connection, voltage outside the rails, or disagreement between the module label and this table.

Specimen gate. The repository has no recorded photograph, label, lot, or revision for the learner’s module. The wiring below is therefore a conditional circuit for the Elegoo EL-SM-001, whose manufacturer product page specifies a 5 V DC supply. If the specimen cannot be matched to that identity and its primary documentation, stop with the bench card open. Do not infer clone ratings from the HC-SR04-shaped board.

Signal	Mega	Connection
Trigger	D40	module Trigger
Echo / TP-E	D41	module Echo and probe tip
Valid evidence	D45	LED plus resistor to GND
Outside evidence	D46	LED plus resistor to GND
Timeout evidence	D47	LED plus resistor to GND
Supply	5 V, GND	module VCC and GND; probe ground to Mega GND



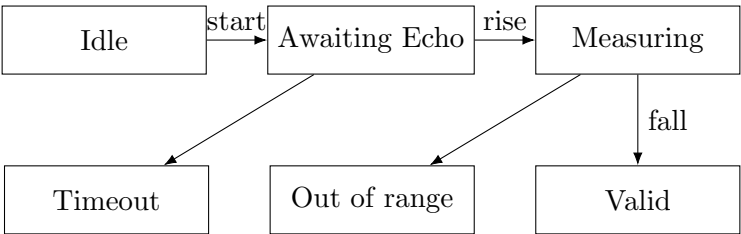
Alternative text: pencil orientation plate showing a Mega, breadboard with three resistor-limited LEDs, ultrasonic module, flat target, Trigger D40, Echo D41/TP-E, and probe ground. The table is authoritative.

3 Deterministic model boundary

UltrasonicRange is deterministic, non-owning, and hardware independent. The sketch’s endpoints own a DigitalOutput for Trigger and a DigitalInput for Echo. Every update supplies both time and the observed Echo level. Resetting the pure model does not release a pin; endpoint shutdown does.

Operation	Contract
initialize()	return Status; validate the range and timing configuration
startMeasurement(now, echoHigh)	return Status; arm from explicit evidence
update(now, echoHigh)	return Status; advance from supplied evidence
reading()	return stable state, distance, Echo duration, and validity
reset()	retain initialization but restore the inert reading state

Check each returned Status with ok(); preserve the complete status when reporting failure. The configured values are 30 ms Echo timeout, 30 ms maximum Echo duration, 20–4000 mm accepted distance, and 343 μm/μs assumed sound speed. These are software classification bounds and a model assumption. They do not certify accuracy, prove a specimen identity, or restate the manufacturer’s full operating envelope.



4 Narrative run loop

```
const adk::MicrosecondTimePoint now (micros ());

if (!observeRange (now))
{
    stopSafely ();
    return;
}

const RangeSignal signal = decideRangeSignal ();

if (!showRangeSignal (signal))
{
    stopSafely ();
    return;
}
```

Objects appear as runtime, endpoints, evidence LEDs, then ranger. Setup acquires them in that order. The three LEDs illuminate together for 150 ms as resource-acquisition evidence. The loop then observes, decides, actuates, and requests a new reading no sooner than every 60 ms. Idle, valid, timeout, and out-of-range are all eligible for that next request; a valid distance cannot stop the measurement cadence and become silently stale.

The 10 μ s Trigger pulse uses one explicit bounded delay. Echo timing and the measurement cadence remain nonblocking. No library behavior allocates, throws, or reads wall-clock time.

5 Predict, observe, interpret

5.1 Experiment A: acquire separately

Predict. D45–D47 illuminate together briefly only after their claims succeed.

Observe: _____.

Interpret. This supports LED and endpoint acquisition; it proves neither Echo timing nor shutdown state.

5.2 Experiment B: Trigger and Echo

Attach probes while unpowered. For a specimen whose primary documentation calls for it, predict a 10 μ s Trigger pulse on D40 and a later high interval on D41/TP-E. Record:

Target	Trigger width	Echo width	LED result
200 mm	_____	_____	_____
500 mm	_____	_____	_____
1000 mm	_____	_____	_____

Estimate distance using

$$d_{\text{mm}} \approx \frac{t_{\text{echo}, \mu\text{s}} \times 343}{2000}.$$

Division by two accounts for the outward and return path. Compare the estimate with the LED category and ruler distance. Do not call this calibration.

5.3 Experiment C: distinct invalid results

Turn the flat target away without touching the powered circuit. A missing accepted Echo should select D47. Move a complete target deliberately outside the documented interval only if the bench permits it; that should select D46. Record what was observed, not what software was expected to do.

5.4 Experiment D: safe state

Invoke the shutdown exercise. D40 must be low before it becomes input/high impedance; TP-E is no longer sampled; all evidence LEDs are inactive. Check Trigger electrically. An LED being off does not prove Trigger safe state or claim release.

6 Diagnosis

Observation	Check next
No startup light walk	endpoint conflict, polarity, resistor, and initialization status
Trigger present; Echo absent	module supply, common ground, target angle, and exact module
Echo present; fault LED	pulse width, timeout boundary, and probe reference
Valid LED; implausible distance	target geometry, Echo width, sound-speed assumption, units
Outside LED at ordinary range	configured limits and integer conversion
Trigger remains driven after stop	safe-state failure; remove USB and record the deviation

7 CLI and bench record

```
make host-test
make arduino-Lesson019UltrasonicRange
make firmware-size EXAMPLE=Lesson019UltrasonicRange
make upload EXAMPLE=Lesson019UltrasonicRange PORT=/dev/ttyACM0
make lessons
make lessons-check
make site-check
```

Use with. The HTML reference carries searchable API notes, corrections, and direct downloads. This PDF is the printable experiment, prediction worksheet, diagnosis table, and acceptance record. Shared safety limits, pins, prerequisites, and API facts appear in both.

- UltrasonicRanger public header and implementation
- PulseInput public header and implementation
- deterministic host tests and canonical Mega 2560 example

Software: commit _____; Arduino CLI _____; AVR core _____; flash _____; RAM _____.

Bench acceptance—open until measured

- ☐ Exact Mega and ultrasonic module recorded.
- ☐ Unpowered wiring and each LED resistor checked.
- ☐ 5 V rail and module supply measured.
- ☐ D40 Trigger width measured.
- ☐ D41/TP-E widths recorded for three ruler distances.

- ☐ Valid, out-of-range, and no-Echo evidence distinguished.
- ☐ Reset, shutdown, resource reuse, and USB removal observed.
- ☐ Trigger safe state checked separately from LEDs.

Supply _____; instrument _____; target _____; date _____; reviewer _____.
 Deviations: _____.

Current result: host verified; hardware acceptance open. Deterministic tests, compilation, and blank worksheets are not physical verification.

8 Primary references

- Elegoo EL-SM-001 product specification: conditional source for the named module's 5 V supply, stated range, acoustic frequency, and model identity.
- Elegoo HC-SR04 documentation index: manufacturer-hosted datasheet route; match it to the physical specimen before bench work.
- Arduino Mega 2560 Rev3 datasheet: official board power and D40–D47 connector identities.

9 Extend the evidence

1. Record how angled cardboard changes Echo validity without changing code.
2. Replay the same timestamp and level trace twice and compare readings.
3. Explain why clamping an invalid distance would erase useful evidence.
4. Design a fourth diagnostic that adds information not already present at TP-E or the three result LEDs.